Fruit Freshness / Ripeness

Group 1

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Objectives

- Detect fruit ripeness automatically using computer vision.
- Train a YOLOv8 Object Detection model.
- Deploy the model for real-time predictions.

Problem

Many fruits look very similar during different ripeness stages and manual inspection:

Takes time

Can be inconsistent

Depends on human judgment

Our goal was to classify fruit ripeness automatically from images.

Methodology / Pipeline

1. Collected & annotated fruit images in Roboflow
2. Exported dataset in YOLOv8-OBB format
3. Trained YOLOv8n (nano) on Colab using a T4 GPU
4. Dataset + trained weights saved to Google Drive, skip re-downloading/re-training automatically
5. Compared 3 hyperparameter configurations, picked the best by mAP50-95

Classes & Dataset

Classes

apple-overripe/ripe/unripe
banana-overripe/ripe/unripe
mango-overripe/ripe/unripe

Dataset

Source: Roboflow
Format: YOLOv8 OBB
Total Images: 813 images

Training / Validation / Test split

592 images / 148 images 73 images

Data Augmentation

Applied in Roboflow before export

Augmentation	Setting
Flip	Horizontal
Brightness	±15%
Blur	Up to 2.5px

Helps the model generalize to different camera angles, lighting conditions, and image sharpness

Model Performance

We trained 3 different models

Run	Epochs	Image Size	Batch	mAP50-95
Baseline	25	512	16	0.9205
More Epochs	50	512	16	0.9237
Bigger Image Size	25	640	8	0.9194

Selected the model with the highest mAP50-95

Final Precision: 99.24%

Final Recall: 98.17%

Final mAP50: 98.59%

Final mAP50-95: 92.37%

Per-Class Performance

Not all classes performed equally

Strongest classes: Apple (all stages) & Mango
mAP50-95 between 0.92–0.99

Weakest class: Banana Unripe, recall as low as 75%
mAP50-95 around 0.78–0.79

Interesting case: Banana Overripe,precision & recall near-perfect (~99–100%), but mAP50-95 only ~0.77–0.81
The model **correctly identifies** it, but the bounding box isn't tightly localized

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Demo

Thank You

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